GROUP WORK PROJECT #1 MScFE 620: Derivative Pricing
GROUP NUMBER: 13486

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. **Note:** You may be required to provide proof of your outreach to non-contributing members upon request.

1. **Put- Call parity- European Options**

   Yes, Put call parity applies for European options. A portfolio of Call+Cash can resemble Put+stock. In Euro. Optn- both Call & put have same K(strike), same T (expiration duration) and exercisable only at expiration. As they settle only @ maturity, final payoff matched.

   @ expiration T, 2 possible scenarios can happen:
   Case 1: St(Stock price) > K then call payoff=St-K and Put =0
   Portfolio A: Call+Present value of K
   Portfolio B: Put+Stock
   @maturity - Portfolio A = (St-K)+K=St
   Portfolio B= 0+St=St
   Thus both equal

   Case 2: St < K
   Call =0 and Put=K- St
   Portfolio A: = 0+K=K & Portfolio B:=(K-St)+St=K

   Since both cases portfolio give same payoff - they must have same price today

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2. **Put-Call parity for call price**

   European Put-call parity is given as :
   $C_0 - P_0 = S_0 - Ke^{-rT}$
   Re-arranging the terms

   $$C_0 = P_0 + S_0 - Ke^{-rT}$$

   This stands that call option is equivalent: Put+Stock - PV of Strike

   We can therefore synthetically create a call option by buying put & stock and risk free borrowing

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3. **Put-Call parity for Put price**

   European Put call parity:

   $$C_0 - P_0 = S_0 - Ke^{-rT}$$

   Solving for Put:

$$-P_0 = S_0 - Ke^{-rT} - C_0$$

$$P_0 = C_0 - S_0 + Ke^{-rT}$$

$$P_0 = C_0 + Ke^{-rT} - S_0$$

Interpretetation : Euro. Put can be formed by : Call + Bond - Stock ie buy call & zero-coupon bond and short the stock.

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4. Put-Call parity for American options

Strict Put/Call parity not hold for American options. However, it do follow inequality bound. The reasons for this is that Amer. option allow early exercise

Americans allow exercise at any time t <= T

Value is

$$\sup_{\tau \leq T} \mathbb{E}^{\mathbb{Q}}[e^{-r\tau}\text{Payoff}]$$

This is supremum over stopping times which eliminate clean payoff as seen in European parity.

Thus the equation $(S_T - K)^+ - (K - S_T)^+ = S_T - K$ only hold at maturity

It lead to inequality version of above equation

$$S_0 - K \leq C_A - P_A \leq S_0 - Ke^{-rT}$$

Where Ca= call price & Pa= Put price

For this reason, it is never optimal to exercise call early but optimal to exercise put early if stock price is low. This flexibility breaks the strict equality of put call parity

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5. Member A (Nhi): DONE

a. I choose the number of steps $N = 500$ for the binomial tree with the given values and parameters:

$$S_0 = 100,\ r = 5\%,\ \sigma = 20\%,\ T = 3\ months,\ K = 100\ (ATM).$$

b. First, I will explain how N affects the achievement of reliable estimates from several aspects.

We know that:

$$dt = T/N$$

In this case, $T = 3\ months = 0.25\ year$ is fixed. As N increases, dt becomes smaller.

At the same time:

$$u = e^{\sigma\sqrt{dt}},\ d = e^{-\sigma\sqrt{dt}}$$

so both $u,\ d$ approach 1. This means that the stock price values in the tree become closer to each other so the tree becomes "smoother".

On the other hand:

$$p = \frac{e^{rdt} - d}{u - d}$$

approaches 0.5, meaning that the probabilities of upward and downward movements become nearly equal. In this case, S_T has a distribution close to log-normal and the option price computed using the binomial tree converges to the price given by the Black-Scholes model.

However, the computational complexity of the binomial tree (which is related to the number of nodes) is approximately $O(N^2)$. Therefore, as N increases, the computational cost increases significantly.

For this reason, I aim to determine a threshold value of N at which the estimated results are sufficiently accurate relative to the computational resources (that is, further increases in N do not materially change the option price).

I test different values $N = 50,\ 100,\ 200,\ 500,\ 1000$ to evaluate this criterion (time represents the computation time). The comparison results are shown in the table:

N	Call	ΔCall	dCall	Put	ΔPut	dPut	Time (s)
25	4.65	0.568171	0.000000	3.41	-0.431829	0.000000	0.001794
50	4.60	0.569120	-0.058734	3.35	-0.430880	-0.058734	0.003871
100	4.61	0.569290	0.009945	3.36	-0.430710	0.009945	0.018276
200	4.61	0.569375	0.004982	3.37	-0.430625	0.004982	0.061012
500	4.61	0.569426	0.002992	3.37	-0.430574	0.002992	0.391701
1000	4.61	0.569443	0.000998	3.37	-0.430557	0.000998	1.566250
2000	4.61	0.569452	0.000499	3.37	-0.430548	0.000499	7.694370
5000	4.61	0.569457	0.000299	3.37	-0.430543	0.000299	45.959255

From the table above, I obtain the following observations:

+) Considering $dCall/dPut$:

- The error decreases approximately at a rate proportional to $1/N$
- After $N = 500$, the change becomes very small
- After $N = 1000$, the change is almost negligible

+) Considering $\Delta Call/\Delta Put$:

- From $N = 500$ onward, the changes are also very small

  +) Considering computation time:

- From $N = 500$ onward, execution time increases significantly, especially from $1000 - 2000$

  Based on these observations, I select $N = 500$ for the binomial tree with the given data.

  The computation process is described as follows:

1. Dividing the maturity T into N equals time steps: $dt = T/N$.
2. At each step, the stock price can move up or down by factors:

$$u = e^{\sigma\sqrt{dt}},\ d = e^{-\sigma\sqrt{dt}}$$

   and risk-neutral probability: $p = \frac{e^{rdt} - d}{u - d}$.

3. Building the stock price tree forward using these up/down movements. At maturity, I compute the payoff of the option at each final node:

$$max(S - K,\ 0) \text{ for a call, } max(K - S,\ 0) \text{ for a put.}$$

4. Moving backward through the tree. At each node, the option value is the discounted expected value of the next step:

$$V = e^{-rdt}(pV_{up} + (1 - p)V_{down}).$$

   Since the option is European, early exercise is not allowed, so I only use the continuation value. The option price is the value at the initial node $V(0,\ 0)$.
5. Δ is calculated using the first step of the tree:

$$\Delta = \frac{V_{up} - V_{down}}{S_{up} - S_{down}}.$$

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6. **Member A (Nhi): DONE**

   I calculated Delta in Question 5 and obtained the results shown in the table below:

N	Call	ΔCall	dCall	Put	ΔPut	dPut	Time (s)
25	4.65	0.568171	0.000000	3.41	-0.431829	0.000000	0.001794
50	4.60	0.569120	-0.058734	3.35	-0.430880	-0.058734	0.003871
100	4.61	0.569290	0.009945	3.36	-0.430710	0.009945	0.018276
200	4.61	0.569375	0.004982	3.37	-0.430625	0.004982	0.061012
500	4.61	0.569426	0.002992	3.37	-0.430574	0.002992	0.391701
1000	4.61	0.569443	0.000998	3.37	-0.430557	0.000998	1.566250
2000	4.61	0.569452	0.000499	3.37	-0.430548	0.000499	7.694370
5000	4.61	0.569457	0.000299	3.37	-0.430543	0.000299	45.959255

a. I compare ΔPut và $\Delta Call$ based on the following criteria: sign (which represents the economic meaning with respect to the stock price), magnitude and the relationship between them.

b. Delta measures the sensitivity of the option price to the underlying stock price. It shows how the option price changes if the stock price increases by 1 unit. With $N = 500$:

- If the stock price increases, the call price also increases, so $\Delta Call > 0$. From the table, it is approximately 0.569. This means that if the stock price increases by 1, the call price increases by about 0.569.

- In contrast, $\Delta Put < 0$ (opposite sign to $\Delta Call$). From the table, it is approximately -0.431. This means that if the stock price increases by 1, the put price decreases by about 0.431.

If we consider the magnitude (ignoring the sign), $\Delta Call$ is larger than ΔPut.

In addition, we can observe that

$$\Delta Call - \Delta Put = 1$$

since the European options do not pay dividends. This is a result of put-call parity.

7.

a. The sensitivity of the option price to volatility is known as Vega. It measures how much the option price changes when volatility changes.

Holding all other parameters constant, volatility (σ) is increased from 20% to 25%. The comparison results are shown in the table above:

Option	Price(20%)	Price(25%)	ΔV	Vega
Call	4.61	5.60	0.9829	19.6582
Put	3.37	4.35	0.9829	19.6582

When volatility increases from 20% to 25%, both the call and put prices increase by the same amount. Specifically:

$$\Delta V_{Call} = \Delta V_{Put} = 0.9829, Vega_{Call} = Vega_{Put} = 19.6528.$$

The positive value of Vega indicates that option prices increase when volatility increases.

b. This result is consistent with theory. From put-call parity:

$C - P = S_0 - Ke^{-rT}$ and the right-hand side does not depend on σ.

Taking the derivative with respect to σ gives: $Vega_{Call} = Vega_{Put}$.

From an economic perspective, option prices are expected to increase when volatility increases, since higher volatility implies greater uncertainty about future stock prices. Recall that:

- $option\ price\ =\ intrinsic\ value\ +\ time\ value$

Because intrinsic value is unaffected by volatility, the increase in option price must come from an increase in time value.

- Moreover, options have asymmetric payoffs. When volatility rises, the probability of extreme outcomes increases. Since options benefit from favorable extreme price movements while limiting losses, they gain value from higher tail risk.

Therefore, the observed increase in both call and put prices is economically reasonable.

8.

N	Call(Am)	Delta(C)	Put(Am)	Delta(P)
50	4.60	0.569120	3.47	-0.450262
100	4.61	0.569290	3.47	-0.449831
200	4.61	0.569375	3.48	-0.449623
500	4.61	0.569426	3.48	-0.449500
1000	4.61	0.569443	3.48	-0.449459

American Option: No of steps & overall process:

- As no of steps N increase 50 to 1000, both option prices gradualize converge indicating stability.
- American Call price stabilized at 4.614 and with Delta 0.5694 confirming no early exercise premium.

- American Put converge to 3.479 with delta at -0.4495
- Comparing step size , the change from = 500 to N= 1000 is negligible
- So N= 500 is recommended as it provide price stability upto 3 decimal and stable Greeks

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9. **American call & Put deltas compare**

N	Delta (Call)	Delta (Put)
50	0.569120	-0.450262
100	0.569290	-0.449831
200	0.569375	-0.449623
500	0.569426	-0.449500
1000	0.569443	-0.449459

Convergence

- As N increase, both delta conver . Call to 0.56944 and Put to -0.44946

Sign Comparison

- Call delta is positive
- Put delta is negative

Magnitude Comparison

- Call > Put --> Thus call more sensitive to stock price change than put

Call delta Positive

- Since payoff = max(S-K,0)

- As stock price increase, call value increase so Delta is positive.
- Delta= 0.569 means if stock increase by 1 unit then call increse by 0.57 units

Put Delta negative

- Since payoff = max(K-S,0)
- stock price incrse, Put decrease. Hence Delta is neg.
- Delta put = -0.449 means if stock increase by 1 unit, put decrease by 0.45 units

Not symmetry

- Becuase American option Put has early exercise value, and hence parity is modifty and hence delta do not sum to 1

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10. **American Option: Sensitivities & Differential impact**
Volatility Sensitivity (20%--> 25%)

Metric	20% Volatility	25% Volatility	Vega
Call Option	4.61300	5.59591	19.65823
Put Option	3.47883	4.45856	19.59467

Interpretation of result on American option due to change in volatility

- CAll price increase significantly. Inc. Prob. of extreme upward moves --> benefit call
- Put price increased significantly. Inc. prob. of extreme downward move --> benefit put.
- $\partial\sigma/\partial V>0$ for both options

Differential impact between call & Put

- Call Vege ~ 19.66 and Put Veg~ 19.60 . 1% increase in volatility increase option price by about 0.196
- This is from put-call parity. Amercial call behave like Euro. call
- Early exercise not optimal. Put has only small early exervise premium

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11. **Member C**

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12. **Member C**

13. **Member C**

14. **Member C**

15. **European Call options Deep OTM, OTM, ATM, ITM,and Deep ITM.**

Moneyness	Strike (K)	Classification	Call Price
90%	90.00	Deep ITM	$11.67
95%	95.00	ITM	$7.72
100%	100.00	ATM	$4.61
105%	105.00	OTM	$2.48
110%	110.00	Deep OTM	$1.19

Observation & Logic:

- As Strike Price (K) increase , Call option price decrease
- Higher price we have to pay to buy stock, less valuable option becomes .
- Deep ITM calls have high intrinsic value while Deep OTM calls rely purely on time value

16. **European Put options Deep OTM, OTM, ATM, ITM,and Deep ITM.**

Moneyness	Strike (K)	Classification	Put Price
90%	90.00	OTM	0.55
95%	95.00	OTM	1.53
100%	100.00	ATM	3.37
105%	105.00	ITM	6.17
110%	110.00	ITM	9.83

Observation & Logic:

- As Strike Price (K) increase , Put option price increase
- Higher price increases the profit potential for right to sell underlying stock

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17.

a. We keep the same input parameters:

$S_0 = 100,\ r = 5\%,\ \sigma = 20\%,\ T = 3\,months = 0.25\,year.$

For an American Call, at each node of the trinomial tree we compute:

$$V = max(Intrinsic\,Value,\ Continuation\,Value)$$

Where:

- $Intrinsic\,Value = max(S - K,\ 0)$
- Continuation Value = discounted expected value under the risk-neutral probability

Since this is an American option, early exercise is allowed. Therefore, at each backward step of the tree, we must compare the intrinsic value with the continuation value. The results are shown in the table below:

Moneyness	Strike (K)	Classification	Call Price
90%	90.00	Deep ITM	$11.67
95%	95.00	ITM	$7.72
100%	100.00	ATM	$4.61
105%	105.00	OTM	$2.48
110%	110.00	Deep OTM	$1.19

b. From the table, it is clear that the Call Price decreases as the Strike (K) increases. This is also consistent with financial theory:

- When K is low (Deep ITM), the Intrinsic Value (a component of the Option Value) is large, so the Call Price is high
- As K increases, the probability that the option finishes ITM decreases
- A Deep OTM option consists almost entirely of Time Value, so the Call Price is low

In this case, since there is no dividend, the $American\ Call\ \approx\ European\ Call$.

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18. Member A (Nhi)

a. Using the same input parameters and the same 5 Strike (K) levels as in the previous question, for an American Put, at each node:

$$V\ =\ max(K\ -\ S,\ Continuation\ Value)$$

The results are presented in the table below:

Moneyness	Strike (K)	Classification	Put Price
90%	90.00	Deep ITM	$0.56
95%	95.00	ITM	$1.57
100%	100.00	ATM	$3.48
105%	105.00	OTM	$6.42
110%	110.00	Deep OTM	$10.33

b. From the table, it can be observed that the Put Price increases as the Strike (K) increases. This is also consistent with financial theory:

- When K is high, the value of K - S is more likely to be large
- The more ITM the Put is, the larger its Intrinsic Value

- A Deep ITM Put has a high probability of payoff, therefore its Put Price is high

Unlike the American Call, a Put option may benefit from early exercise (especially when it is deep ITM). Therefore, in general, the $American\ Put\ \geq\ European\ Put$.

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19. **Member C**

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20. **Member C**

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21. **Member C**

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22. **Member C**

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23. **Member C**

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24. **Member C**

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25. **Dynamic Delta Hedging : European Put Option**

3-Step Binomial Tree (European Put)

Parameters
S0 = 180
u = 1.1075
d = 0.9030
K = 182

```
 Binomial Tree (3 Steps) For European Put Option:
Stock Prices:
 [[180.   199.34 220.76 244.48]
 [  0.   162.54 180.   199.34]
 [  0.     0.   146.77 162.54]
 [  0.     0.     0.   132.52]]
Option Values:
 [[13.82  5.01  0.    0.  ]
 [ 0.   22.41  9.88  0.  ]
 [ 0.    0.   34.63 19.46]
 [ 0.    0.    0.   49.48]]
Deltas:
 [[-0.4726 -0.2423  0.    ]
 [ 0.     -0.7447 -0.5288]
 [ 0.      0.     -1.    ]]
p: 0.4908, u: 1.1075, d: 0.9030
```

Acting as seller (writer) of the put we receive **Put price = 13.82** at t=0
We dynamically hedge using delta. Since delta is negative, we will short stock.

<u>3 steps Binomial Tree path</u>

t=0: 180

t=1: 199.34 162.54

t=2: 220.76 180 146.77

t=3: 244.48 199.34 162.54 132.52

We choose path Up→Down→Down
Ie.
t=0 180| t=1 199.34|t=2 180|t=3 162.54
Final payoff: max(182-162.54,0) = 19.46

<u>Step-by-step delta hedging</u>:
Step 0(t=0) delta = -0.4726 -> since we are short the put→ hedge by shorting 0.4726 shares
Stock proceeds received: .4726 * 180 = 85.07
Initial cash: 13.82+85.07 =98.89

Step 1(Up move → S=199.34)
New delta = -0.2423 → currently short 0.4726 shares . New hedge require to short 0.2423 shares. So we net we buy back: 0.4726 - 0.2424 = 0.2303
Cost: 0.2303*199.34 = 45.90
Cash before rebalance grows at risk free rate : 98.89 *e^(0.02*⅙)*=99.22
After buying shares : 99.22 -45.90 = 53.32

Step 2(Down move → S = 180)
New delta = -0.7447
Currently short 0.2423 Need to short 0.7447. So additional short: 0.7447 - 0.2423= 0.5024
Cash received: 0.5024 * 180 = 90.43
Cash grow: 53.32 * e^(0.02*⅙)=53.50
After shorting: 53.50+90.43= 143.93

Step 3 (Down move-> S=162.54)
Delta = -1
Currently short 0.7447
Need to short additional: 1-0.7447 = 0.2553
Cash received: 0.2553 * 162.54 = 41.49
Cash grows: 143.93*e^(0.02*⅙)= 144.12
After shorting : 144.12+41.49=185.61

@ maturity: we are short 1 share:
To close hedge- we buy 1 share at 162.54 thus
Cash: 185.61-162.54=23.07
Put option payoff: 23.07-19.46=3.61

Time	Stock	Delta	Shares Short	Cash Account
t0	180	-0.4726	-0.4726	98.89
t1 (U)	199.34	-0.2423	-0.2423	53.32
t2 (D)	180	-0.7447	-0.7447	143.93
t3 (D)	162.54	-1.000	-1.000	185.61

26.

Dynamic Delta Hedging : American Put Option
Using the given S0=180, K=182, r=2%,sigma=25%, T=6 months and Steps= 25

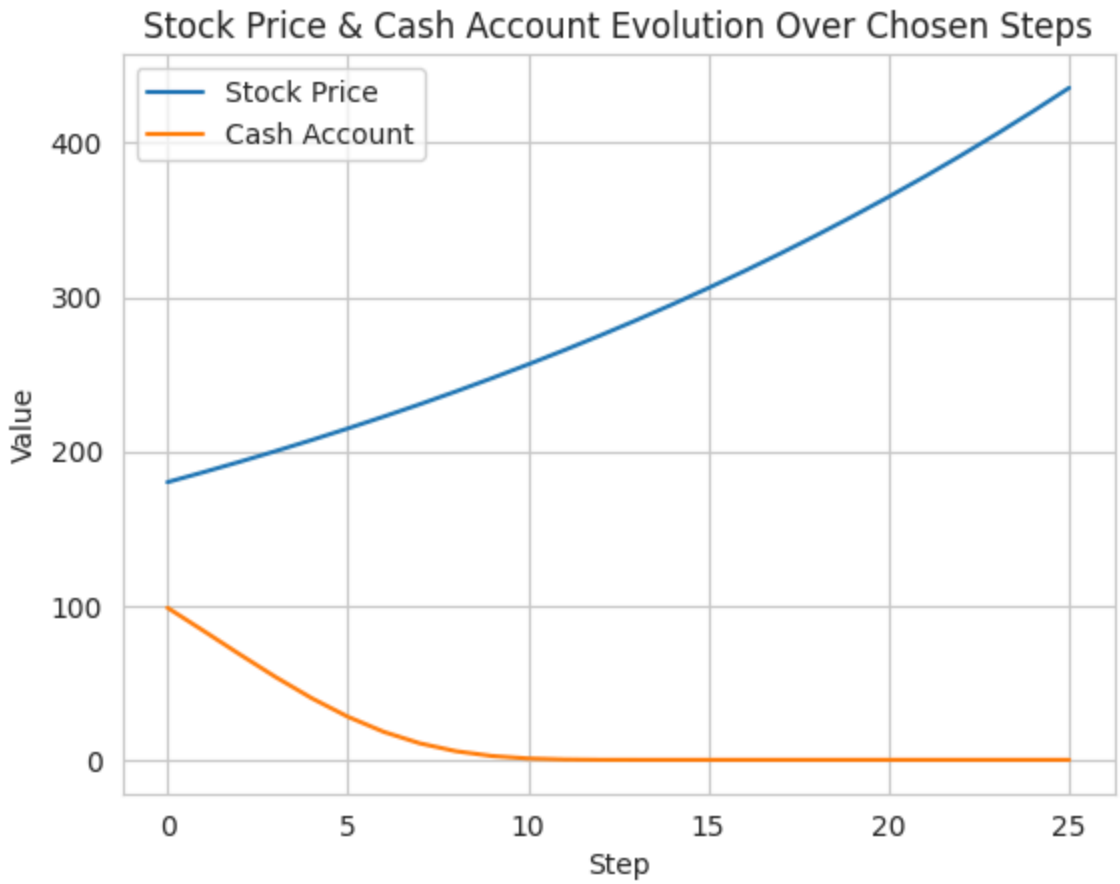
When using the binomial tree model with 25 steps, we get

Calculated American Put Option Price : 13.04
Delta = -0.4756

The American Put prices feed in opportunity for early exercise @ each node in tree.

Simulation Path (all UP &UP): Following is the delta hedging required and cash account

Step	Stock Price	Put Value	Delta	Cash Account
0	180.00	13.04	-0.4756	98.64
1	186.48	9.99	-0.3951	83.67
2	193.19	7.38	-0.3163	68.48
3	200.14	5.20	-0.2423	53.69
4	207.34	3.48	-0.1760	39.98
5	214.81	2.18	-0.1200	27.97
6	222.54	1.27	-0.0758	18.13
7	230.54	0.67	-0.0435	10.69
8	238.84	0.31	-0.0221	5.59
9	247.44	0.12	-0.0096	2.49
10	256.34	0.04	-0.0033	0.89
11	265.57	0.01	-0.0008	0.23
12	275.12	0.00	-0.0001	0.03
13	285.02	0.00	0.0000	0.00
14	295.28	0.00	0.0000	0.00
15	305.91	0.00	0.0000	0.00
16	316.92	0.00	0.0000	0.00
17	328.32	0.00	0.0000	0.00
18	340.14	0.00	0.0000	0.00
19	352.38	0.00	0.0000	0.00
20	365.06	0.00	0.0000	0.00
21	378.20	0.00	0.0000	0.00
22	391.81	0.00	0.0000	0.00
23	405.91	0.00	0.0000	0.00
24	420.52	0.00	0.0000	0.00
25	435.65	0.00	0.0000	0.00



Comparison to the European Option case

- Fundamental difference is early exercise in American
- Initial Delta for Amer= -0.4756 incorporates early exercise while Europe delta= -0.4726 based on expectation of exercise at maturity
- Delta more sensitive to change in stock price in Americas than European
- Hedging becomes complex in American due to frequent rebalancing and higher transaction cost
- Delta diverge more signification when the option goes in Deep in money because of early exercise flexibility in American

27. Asian ATM Put

Unlike American put, the Asia Put payoff is calculated using the average price (S bar).

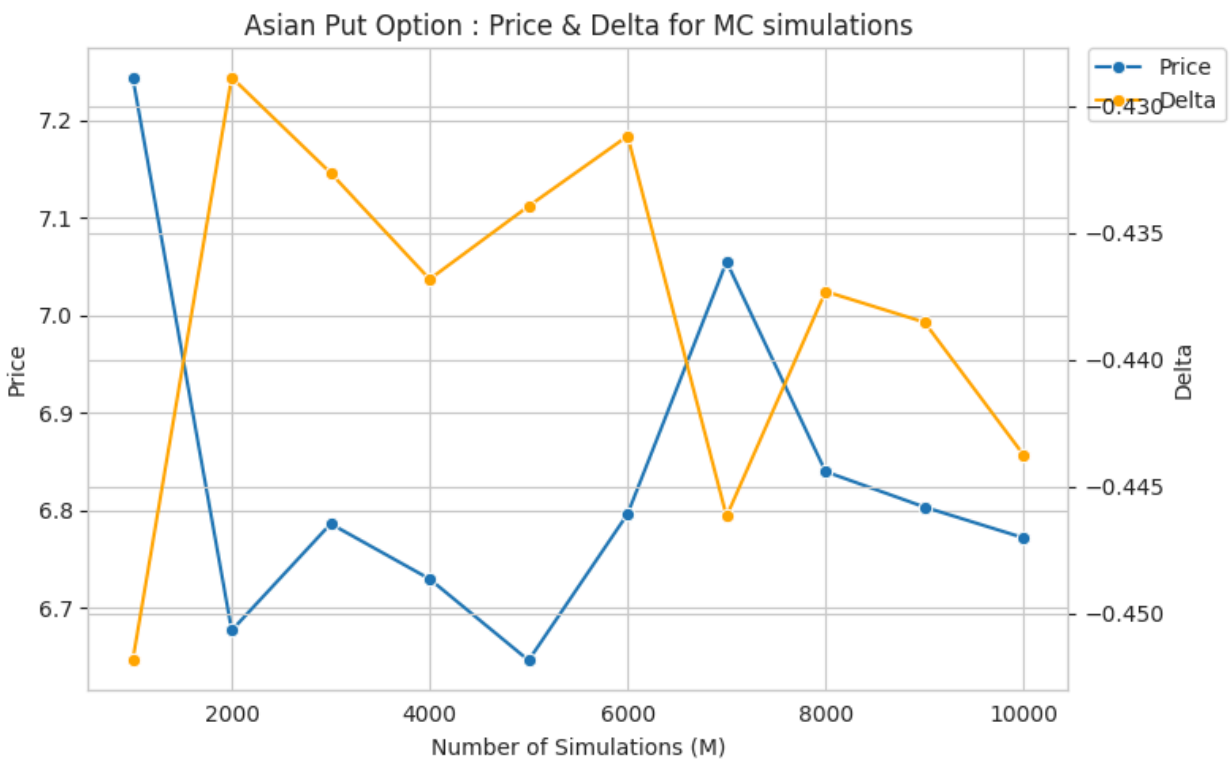
Payoff= **max(K-**$\bar{S}$**,0)** where $\bar{S}$ **= 1/n**ΣSi
ATM: K=So=180

Methodology:

1. Simulated multiple price paths
2. Calculated avg price for each path
3. Discount avg payoff to present.
4. Calculated Delta

M	Price	Delta
1000	7.11	-0.4593
2000	6.55	-0.4355
3000	7.04	-0.4561
4000	6.87	-0.4406
5000	6.84	-0.4445
6000	6.89	-0.4415
7000	6.85	-0.4440
8000	6.80	-0.4415
9000	6.83	-0.4458

10000	**6.88**	**-0.4457**



Comparison Observations with American Put:

1. Price generally lower because averaging reduce volatility
2. As seen below- spikes are smooth out by average
3. Delta is more stable as average becomes harder to change

Comment on Delta hedgin process for Asia Put Option compared to European Put

- Asian Put is significantly complexer than Europen due to the average payoff in Asia instead of spot
- Europen Put- delta sensite to spot move until end. But in Asia, average becomes very resilient to change.
- Europen is driven by Proability St<K but in Asia delta is driven by I/T Sum(Stdt)<K

Asian Put option Evolution of cash account through chosen step (all up & up)

- We notice that Delta starts around -0.4104 and rapidly approached 0. Since the stock price is rising, probability of price staying under 180 vanishes
- Cash account movement: we are selling Put, we start with premium, , As delta moves from neg towards 0, we are buying stocks ,at higher and higher prices, which depletes cash.
- Compared to Eurpoean & American- Delta changes are more smooth because average price is less volatile.
- This leads to Lower Gamma Risk
- In all up Path, asia put becomes worthless.

Final Average Price: 15.05

Final Payoff: 84.95

Final Portfolio P&L: -122.7425

Step	Stock_Price	Delta	Cash_Account

0	**180.00**	**-0.4674**	**53.47**
1	**186.48**	**-0.2147**	**6.41**
2	**193.19**	**-0.1349**	**-9.00**
3	**200.14**	**-0.1193**	**-12.13**
4	**207.34**	**-0.0071**	**-35.42**
5	**214.81**	**-0.0068**	**-35.50**
6	**222.54**	**-0.0023**	**-36.55**
7	**230.54**	**-0.0006**	**-36.97**
8	**238.84**	**0.0000**	**-37.15**
9	**247.44**	**0.0000**	**-37.19**
10	**256.34**	**0.0000**	**-37.23**
...	**...**	**...**	**...**

20	**365.06**	**0.0000**	**-37.60**
25	**435.65**	**0.0000**	**-37.79**

References:

- Hull, J. C. (2021). *Options futures and other derivatives*. 11th Edition. Pearson. Chapter 13.